An element-assignment computation of the number of extremal even unimodular lattices of rank 32

Auxiliary note to "The extremal even unimodular lattices of rank 32"

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This note records an organization of the count by assigned group elements: every automorphism other than ± 1 is assigned to one element of a short list, and the count becomes a sum of thirteen terms involving the numbers $e(h)$ and the masses $M_2, ..., M_5$. It was used as a check of the classification-based proof of the paper and is not a premise of that proof. It keeps the arrangement of an earlier version of the paper: Sections 3–5, Tables 5–7 and Appendix C.8 of this note are its own, and its references to the paper use that arrangement. There, Lemma 2.4, Propositions 3.4 and 3.5, Lemmas 4.3, 4.4 and 4.5, Theorem 4.6 and Proposition 4.7 are Lemma 2.2, Propositions 3.2 and 3.3, Lemmas 4.2, 4.3 and 4.4, Theorem 4.5 and Proposition 4.6 of the paper, and Tables 4 and 4a are Tables 5 and 6. Its sections keep their numbers in that arrangement, so the numbering has gaps where sections are now in the paper (§4.1.1 follows §3.2), and a reference to a section that this note does not contain says "of the paper". The table of its Appendix C.8 is called Table C8 here so as not to be confused with Table C2 of the paper. Definitions and references are those of the paper.

3. The counting formula

What this section does. We rewrite N as a sum over automorphisms (§3.1), assign every automorphism $g \neq \pm 1$ to exactly one element of a small list (Proposition 3.2), and obtain formula (3.3), which involves no individual $|\mathrm{Aut}\ L|$. We then list the possible types of the assigned elements (§3.3 of the paper and its Table 4). Everything in this section is elementary, except that Proposition 3.5 uses the exact linear programs and lists of [C2]; its proof also uses Lemma 4.5 from §4.2 of the paper, which does not depend on §3.

3.1 Summing over automorphisms

A *pair* (L, h) consists of $L \in \mathcal{L}$ and $h \in \mathrm{Aut}\ L$; pairs (L, h) and (L', h') are isometric if there is an isometry $u : L \to L'$ with $uhu^{-1} = h'$. For fixed L, the isometry classes of pairs (L, h) are the conjugacy classes of $\mathrm{Aut}\ L$, and the class of h has $|\mathrm{Aut}\ L| \ / \ |C(h)|$ elements. Since $g \mapsto 1$ summed over $\mathrm{Aut}\ L$ gives $|\mathrm{Aut}\ L|$, and $g = 1$, $g = -1$ contribute $2/\ |\mathrm{Aut}\ L|$ to each L,

$$N = \sum_{L \in \mathcal{L}} \frac{1}{|\mathrm{Aut}\ L|} \sum_{g \in \mathrm{Aut}\ L} 1 = 2m + \sum_{L \in \mathcal{L}} \frac{\#\{g \in \mathrm{Aut}\ L : g \neq \pm 1\}}{|\mathrm{Aut}\ L|} \tag{3.1}$$

and, since $|\mathrm{Aut}\ L| = 2$ exactly when $L \notin S$,

$$N = 2m + \sum_{L \in S} \left(1 - \frac{2}{|\mathrm{Aut}\ L|}\right). \tag{3.2}$$

As $|\mathrm{Aut}\ L|$ is even, each term of (3.2) lies in $[1/2, 1)$, so $2m + |S| \ /2 \leq N < 2m + |S|$. Formula (3.2) needs the individual $|\mathrm{Aut}\ L|$; formula (3.3) below does not.

3.2 Assignment of automorphisms

Proposition 3.2. Let $g \in \mathrm{Aut}\ L$, $g \neq \pm 1$, of order n. Exactly one of the following holds.

1. n is divisible by an odd prime; let p be the smallest one; then $h = g^{n/p}$ has order p and $g \in C(h)$.
2. $n = 2$: g is an involution $\sigma \neq \pm 1$.
3. $n = 2^a$ with $a \geq 2$ and $\sigma = g^{2^{a-1}} \neq -1$: σ is an involution $\neq \pm 1$ and $g \in C(\sigma)$.
4. $n = 2^a$ with $a \geq 2$ and $g^{2^{a-1}} = -1$: the minimal polynomial of g is Φ_{2^a}, and L is a Hermitian $\mathbb{Z}[\zeta_{2^a}]$-lattice of rank $32/2^{a-1}$; moreover $a \leq 5$.

Proof. Since $g \neq 1$, $n > 1$. If n has an odd prime factor we are in case 1, and $h = g^{n/p}$ has order p and commutes with g. Otherwise $n = 2^a$ with $a \geq 1$. If $a = 1$, g is an involution, different from ± 1 by assumption: case 2. If $a \geq 2$, $\sigma = g^{2^{a-1}}$ has order 2. Either $\sigma = -1$ (case 4) or $\sigma \neq -1$; in the latter case $\sigma \neq 1$ as well, so σ is an involution $\neq \pm 1$ commuting with g (case 3). The cases are distinguished by n and by whether $g^{n/2} = -1$, so exactly one holds. In case 4, g is a root of $X^{2^{a-1}} + 1 = \Phi_{2^a}$, which is irreducible over $\mathbb{Q}$, so it is the minimal polynomial of g; then $a \leq 5$ by Lemma 2.4 and the remark after it. □

We say that g is *assigned* to h in case 1, to $\sigma = g$ in case 2 and to σ in case 3. An element assigned to h lies in $C(h)$, and assignment commutes with conjugation: if g is assigned to h, then ugu^{-1} is assigned to uhu^{-1} for every $u \in \mathrm{Aut}\ L$. For h of odd prime order, or an involution $h \neq \pm 1$, let $e(h)$ be the number of elements of $\mathrm{Aut}\ L$ assigned to h; it depends only on the isometry class of (L, h), and $e(h) \geq 1$ since h is assigned to itself. For $a = 2, ..., 5$ let

$$M_a = \sum_{(L,g)} \frac{1}{|C(g)|}, \tag{3.3a}$$

the sum over isometry classes of pairs with $L \in \mathcal{L}$ and $g^{2^{a-1}} = -1$. Each M_a is a mass of rootless Hermitian lattices over $\mathbb{Z}[\zeta_{2^a}]$: for $a = 2$ by §4.3 of the paper, for $a = 3, 4, 5$ by Appendix C.8 of this note.

Corollary 3.3.

$$N = 2m + \sum_{(L,h)} \frac{e(h)}{|C(h)|} + \sum_{a=2}^{5} M_a, \tag{3.3}$$

where (L, h) runs over isometry classes of pairs with $L \in \mathcal{L}$ and h of odd prime order or an involution $\neq \pm 1$.

Proof. Fix L and put $G = \mathrm{Aut}\ L$. By Proposition 3.2, $G \setminus \{\pm 1\}$ is the disjoint union of the sets $A(h)$ of elements assigned to h, over the elements h of odd prime order and the involutions $h \neq \pm 1$, and of the sets $B_a = \left\{g : g^{2^{a-1}} = -1\right\}$ for $2 \leq a \leq 5$. Grouping the h into conjugacy classes, a class of size $|G| \,/\, |C(h)|$ contributes $(|G| \,/\, |C(h)|) \cdot e(h)$ elements; grouping B_a into conjugacy classes gives $|B_a| = \sum_{[g]} |G| \,/\, |C(g)|$. Divide by $|G|$, sum over $L \in \mathcal{L}$ and insert in (3.1). □

No individual $|\mathrm{Aut}\ L|$ occurs in (3.3): only the centralizers of the elements h and g, and the numbers $e(h)$, which are computed inside $C(h)$.

4.1.1 Orientation

For an involution, σ and $-\sigma$ are different elements, and the pair $(L, -\sigma)$ has fixed lattice K. If $F \nsim K$, the pair $(L, -\sigma)$ appears in the list of type $(32 - d, d)$ as the class of φ^{-1}, not in the list of type $(d, 32 - d)$; we compute both contributions on the list of $(d, 32 - d)$, and the contribution of the type is $E =$

$E(\sigma)+E(-\sigma)$ with $E(\pm\sigma)=\sum_{[\varphi]} e(\pm\sigma)/\,|\mathrm{Stab}(\varphi)|$. If $F\cong K$ (type $(16,16)$ with equal components), the classes $[\varphi]$ and $[\varphi^{-1}]$ in the same list correspond to (L,σ) and $(L,-\sigma)$, and the contribution is $E=E(\sigma)$ summed over all classes of the list; the two coincide exactly when $\sigma\sim-\sigma$. In both cases E is the sum of $e(\sigma)/\,|\mathrm{Stab}(\varphi)|$ over the extremal classes $[\varphi]$ of the list, together with that of $e(-\sigma)/\,|\mathrm{Stab}(\varphi)|$ when $F\nsim K$.

Lemma 4.2. Let $\sigma\neq\pm1$ be an involution and g an element of order 2^a, $a\geq 2$, with $g^{2^{a-1}}=\sigma$. Then 2^{a-1} divides rank K. In particular $e(\sigma)=1$ if rank K is odd.

Proof. g commutes with σ, so it preserves $K\otimes\mathbb{Q}$, on which $g^{2^{a-1}}=-1$. As in Proposition 3.2, the minimal polynomial of $g|_K$ is the irreducible polynomial Φ_{2^a} of degree 2^{a-1}, so 2^{a-1} divides rank K. Elements assigned to σ other than σ itself are of this form (case 3 of Proposition 3.2). If rank K is odd, no such g exists, since $2^{a-1}\geq 2$; then σ is the only element assigned to σ, and $e(\sigma)=1$. □

For types $(15,17)$ and $(17,15)$ both ranks are odd, so $e=1$ and the contribution is twice the extremal mass of the list of type $(15,17)$.

The numbers $e(\sigma)$. If g is assigned to σ in Proposition 3.2, then σ is a power of g, so g commutes with σ; hence $e(\sigma)$ is the number of $g\in C(\sigma)$ with $g=\sigma$ or $g^{2^{a-1}}=\sigma$ for some $a\geq 2$. For each extremal class, generators of $\mathrm{Stab}(\varphi)\cong C(\sigma)$ are lifted to automorphisms of L_φ (checked in integers), and $e(\sigma)$ and $e(-\sigma)$ are counted in this group with GAP [C34]. By Lemma 4.2 only a with $2^{a-1}\mid 32-d$ occur for σ, and only a with $2^{a-1}\mid d$ for $-\sigma$.

The numbers $e(h)$. An element g is assigned to h when the smallest odd prime dividing its order n is p and $g^{n/p}=h$; then $g\in C(h)$. If $p^2\nmid|C(h)|$, the Sylow p-subgroup $\langle h\rangle$ is central in $C(h)$, so $C(h)=\langle h\rangle\times H$ by the Schur–Zassenhaus theorem, and writing $g=h^at$ with $t\in H$ of order m one finds that g is assigned to h if and only if m has no odd prime factor smaller than p and $am\equiv 1\pmod p$. Hence

$$e(h)=\#\{t\in H:\text{the order of } t \text{ has no odd prime factor} < p\},$$

and, under the same hypothesis $p^2\nmid|C(h)|$, $e(h)=|C(h)|\ /3$ for $p=3$. In general $e(h)$ was counted in $C(h)$, built from generators of $\mathrm{Stab}(\varphi)$ lifted to L_φ and checked in integers, by conjugacy classes in GAP [C8, C9]; in all $1{,}234+204$ cases where the group was built, its order equals $|\mathrm{Stab}(\varphi)|$.

Table 5. Elements of odd prime order: the contribution $E=\sum e(h)/\,|C(h)|$ of each type (exact) [C8, C9].

type	E	approx.
$3-(9,14)-7$	3,302 / 6,561	0.503
$3-(10,12)-6$	10/81	0.123
$3-(10,12)-8$	7/3	2.333
$3-(8,16)-8$	1,438 /81	17.753
$3-(12,8)-8$	4,187,858 / 2,187	1,914.887
$3-(9,14)-9$	28,961 /729	39.727
$3-(10,12)-10$	9,640 /81	119.012
$3-(13,6)-5$	3,676 /243	15.128
$3-(6,20)-6$	10,048 / 2,187	4.594
$3-(16,0)-0$	413/243	1.700
order 3	**13,881,509 / 6,561**	**2,115.761**
$5-(4,16)-4$	38/ 1,875	0.020

type	E	approx.
$5-(5,12)-5$	214/625	0.342
$5-(6,8)-4$	296/625	0.474
$5-(6,8)-6$	442/15	29.467
$5-(7,4)-3$	836/625	1.338
$5-(8,0)-0$	8,192 / 16,875	0.485
$7-(4,8)-4$	134/63	2.127
$17-(2,0)-0$	13/51	0.255
$31-(1,2)-1$	8/31	0.258
orders 5, 7, 17, 31	**2,164,244,689 / 62,251,875**	**34.766**

The enumeration also confirms Theorem 4.6. Its sum is a lower bound for M_2 whatever the formula says, and equality could hide an error only if an error in the formula were exactly compensated by missing lattices of the same total weight. For $a = 4$ an independent list of 514 classes, of which 238 are rootless, has total weight M_4. For $a = 3$ (rank 8) no independent list exists; the value rests on the proof of the formula and on its agreement at ranks 2, 4 and 6 (Table 6).

Table 6. Rootless Hermitian masses M_a.

a (rank)	M_a	approx.	confirmed by
2 (16)	4,128,748,922,231,417 / 120,324,096,000	34,313.567	enumeration of 76,742 classes, exact [C12]
3 (8)	1,373,459,063 / 7,741,440	177.416	no independent check at rank 8; the same formula agrees at ranks 2, 4, 6, and at rank 6 the rootless value 1/ 48,384 agrees with a centralizer order in $2.\mathrm{Co}_1$ [C35]
4 (4)	6,983 /640	10.911	list of 514 classes [C35]
5 (2)	31/24	1.292	explicit count of the 16 types of rank 2 (Table C8)

4.4 The thirteen terms

Table 7. The thirteen terms of (3.3) [C36]. The term $2m$ is twice (2.1).

term	exact value	approx.	§
$2m$	1,310,037,331,282,023,326,658,917 / 119,431,715,880,960,000	10,968,923.302	2.2
M_2	4,128,748,922,231,417 / 120,324,096,000	34,313.567	4.3
$(16,16)$, $s=16$, $\delta=1$	2,880,305,926,221 / 293,601,280	9,810.264	4.1
$(15,17)$	20,874,607 / 2,160	9,664.170	4.1
$(14,18)$	491,776,351,403 / 70,778,880	6,948.066	4.1
$(8,24)$	4,903,343,345,383,327 / 792,723,456,000	6,185.440	4.1
$(12,20)$	24,698,881,091,413 / 4,246,732,800	5,815.972	4.1
order 3	13,881,509 / 6,561	2,115.761	4.2
$(16,16)$, other 10 pairs	1,524,348,533,472,859 / 5,707,608,883,200	267.073	4.1
M_3	1,373,459,063 / 7,741,440	177.416	C.8
orders 5, 7, 17, 31	2,164,244,689 / 62,251,875	34.766	4.2
M_4	6,983 /640	10.911	C.8
M_5	31/24	1.292	C.8

term	exact value	approx.	§
sum	**11,044,268**		5

4.5 One term in full: $31-(1,2)-1$

Every term of Table 7 for an element of odd prime order or an involution is made in the same way: list the components, list their gluings, keep the extremal ones, and count the elements assigned to h. For most types the lists are long and the steps are done by programs. For the type $31-(1,2)-1$ every step can be done by hand, and we do so here; the result is the entry 8/31 of Table 5.

Let h be an automorphism of order 31 of $L\in\mathcal{L}$ of type $31-(1,2)-1$. Then $F=L^h$ has rank 2 and $F^\#/F\cong\mathbb{F}_{31}$, and $Z=F^\perp\cap L$ has rank 30 and is a Hermitian $\mathbb{Z}[\zeta_{31}]$-lattice of rank 1 with $Z^\#/Z\cong\mathbb{F}_{31}$ (§3.3 of the paper).

The lattice F. F is a positive definite even binary lattice of determinant 31, with Gram matrix $\left(\begin{smallmatrix}2a & b\\ b & 2c\end{smallmatrix}\right)$ and $b^2-4ac=-31$. By the reduction of binary forms, the reduced forms $ax^2+bxy+cy^2$ with $|b|\le a\le c$ are $(a,b,c)=(1,1,8)$ and $(2,\pm1,4)$, and the last two give isometric lattices. Hence the genus consists of two lattices, $\left(\begin{smallmatrix}2 & 1\\ 1 & 16\end{smallmatrix}\right)$ of minimum 2 and

$$F=\begin{pmatrix}4 & 1\\ 1 & 8\end{pmatrix},$$

which is the only one of minimum at least 4. Since $0<b<a<c$, $\mathrm{Aut}\,F=\{\pm1\}$. The dual $F^\#$ has Gram matrix $\frac{1}{31}\left(\begin{smallmatrix}8 & -1\\ -1 & 4\end{smallmatrix}\right)$, so $\min F^\#=4/31$; in particular $F^\#$ has no vector of norm 2/31. On $F^\#/F\cong\mathbb{F}_{31}$ the discriminant form is non-degenerate of dimension 1, so $O(q_F)=\{\pm1\}$, and $\mathrm{Aut}\,F\to O(q_F)$ is onto.

The lattices Z. Here $s=z=1$, so Z is unimodular for the form of Lemma 4.5, and Z runs over the genus $(1,1)$ of Proposition D.1, the rank-1 Hermitian lattices, that is, ideal classes with their forms. This genus has $h^-(\mathbb{Q}(\zeta_{31}))=9$ classes: the minus class group of $\mathbb{Q}(\zeta_{31})$ is cyclic of order 9, a result of Kummer [Sch98, p. 1226]. Independently, the mass of the genus given by the formula of Appendix D is 9/62 [C2], which with $|\mathrm{U}(Z)|=62$ for every class (below) again gives nine classes. A unitary automorphism of a lattice of rank 1 is multiplication by a unit u with $u\overline{u}=1$; all conjugates of such a u have absolute value 1, so u is a root of unity, and $\mathrm{U}(Z)=\langle-\zeta_{31}\rangle$ has order 62 for every class. By (4.2), a root x of Z spans an orthogonal summand $\mathbb{Z}[\zeta_{31}]x\cong A_{30}$, which for rank 1 is all of Z; so exactly one class, A_{30}, has roots. The other eight classes are rootless, with total weight $8/62=4/31$, the entry of Table D1; being even and rootless, they have minimum at least 4.

The gluings. $\mathrm{Iso}(q_F,-q_Z)$ has $|O(q_F)|=2$ elements. The image of $\mathrm{Aut}\,F$ in $O(q_F)=\{\pm1\}$ is everything, and so is the image of $\mathrm{U}(Z)$, because ζ_{31} acts trivially on $Z^\#/Z$ (proof of Proposition 4.1(i)) and -1 acts as -1. So for each Z there is exactly one double coset, with

$$\mathrm{Stab}(\varphi)=\left\{\left(\varepsilon,\varepsilon\zeta_{31}^k\right):\varepsilon=\pm1,0\le k<31\right\}$$

of order 62, in agreement with (4.1): $1/62=2/(2\cdot62)$. By Lemma 4.4, since $F^\#$ has no vector of norm 2/31, every gluing is extremal. This gives eight pairs (L,h), one for each rootless Z.

The centralizer and $e(h)$. By Proposition 4.1(iii), $C(h)\cong\mathrm{Stab}(\varphi)=\langle-1\rangle\times\langle h\rangle$, cyclic of order 62. Since $31^2\nmid62$, the formula of the paragraph *The numbers* $e(h)$ above gives $e(h)=\#\{t\in\{\pm1\}\}=2$: the elements assigned to h are h itself and $g=-h^{16}$, of order 62 with $g^2=h^{32}=h$.

The contribution. Each of the eight pairs contributes $e(h)/\,|C(h)|=2/62$, so the type contributes

$$E = 8 \cdot \frac{2}{62} = \frac{8}{31} \approx 0.258,$$

the entry of Table 5. Integrality of the whole sum (§5.3) would detect an error here: a ninth rootless class would change E by $2/62 = 1/31$, and an error of one in one $e(h)$ would change it by $1/62$.

The underlying lattices of these eight pairs need not be pairwise non-isometric and are not needed for this count. They can nevertheless be constructed. Taking for Z the ideal $P/\overline{P}$ for a prime P above a prime $\ell \equiv 1 \pmod{31}$, the fourteen primes $\ell < 3{,}200$ give lattices Z of minimum 4, and the lattices L obtained have the invariants of BW_{32} or of the code lattice of the extended quadratic residue code of length 32 (Appendix A); this sample is not used in the proof [C9].

5.3 What integrality checks

That the sum is an integer is not used in the proof, but it detects many errors. A pair (L, h) contributes $e(h)/\ |C(h)|$ with $1 \le e(h) \le |C(h)| - 1$, since $1 \in C(h)$ is never assigned to h. Hence an error in a single $e(h)$, or a single pair wrongly kept or discarded as extremal, changes the sum by a non-zero number of absolute value less than 1, which makes it non-integral. Errors that are not detected are: two errors that cancel modulo 1; an error by an integer amount in one term, which is possible only for the masses $2m$, M_a and the totals of large lists; and a uniform error of a whole term by an integer factor. Against these the proof relies on §4 and Appendices B–D, not on integrality.

5.4 How each term is checked

Every value in Table 7 is protected in three ways, and Table 7a collects them term by term.

1. *Completeness within a term.* No list in §4 is trusted because of how it was found. Each one is complete by an exact equality of masses (Lemma 7.1): (4.1) for the double cosets of each of the 83 pairs of components of involutions and of each pair of odd prime order; (B.1) for the lattices K of type $(8, 24)$; the mass of the genus, the rootless mass and (B.2) for the components of odd prime order (Table B4). A missing class would leave the left side short of the right side, unless the value on the right were wrong by exactly the missing weight. The components of involutions are complete instead by the exhaustive enumeration of Lemma B.1.
2. *Independent confirmations.* A value is confirmed when a second, independent method gives the same fraction: King's table for m, the enumeration of Proposition 4.7 for M_2, published classifications for two types, and the checks of Appendices B.2, C.7 and D.3. For the pairs with $e(h) > 1$ of the types $(16, 16)$ with $\delta = 1$, $(14, 18)$, $(12, 20)$ and $(8, 24)$, $e(h)$ was also recounted inside $\mathrm{Aut}\ L$ computed with bliss, for all pairs with $|C(h)| \le 64$ and for all 152 pairs of type $(16, 16)$, $\delta = 1$ with $|C(h)| > 64$, with no disagreement; these recounts use the same counting program and check the lifting of the groups, not the counting; the counting programs were checked in rank 24 against Co_0 (§5.6), which again checks the programs, not the values in rank 32.
3. *Global identities.* The integer auxiliary sum and the historical element-by-element comparisons test the assignment computation. They share component lists with the classification and do not replace the completeness or full-group arguments. For the main proof, (5.1) is aggregated directly from the classified rows [C13].

Table 7a. How each term of Table 7 is computed and checked. "Complete by" is the identity that proves its lists complete; "confirmed by" lists independent checks of its value. The global identities are in the third item above the table; what remains assumed is in Table 10 of the paper.

term	computed by	complete by	confirmed by
$2m$	King's inclusion–exclusion [C14]	— (a formula)	King's table [Kin03], exactly
M_2	formula of App. C [C11]	— (a formula)	enumeration of 76,742 classes (Prop. 4.7) [C12]
$(16,16)$, $s=16$, $\delta=1$	gluings [C4]; e [C34]	(4.1); components by Lemma B.1	components by a classification of the genera (§B.2); e recounted in Aut L (lifting only, §5.4); the program for e in rank 24 (§5.6)
$(15,17)$	gluings [C4]; $e=1$ (Lemma 4.2)	(4.1); components by Lemma B.1	rootless masses of the genera of the components (§B.2)
$(14,18)$	gluings [C4]; e [C34]	(4.1); components by Lemma B.1	rootless masses (§B.2); e recounted for $\lvert C\rvert \le 64$; the program for e in rank 24 (§5.6)
$(8,24)$	Theorem B.3, Niemeier lattices [C6]; e [C34]	(B.1) for Leech, A_1^{24}, A_2^{12}	e recounted for $\lvert C\rvert \le 64$; the program for e in rank 24 (§5.6)
$(12,20)$	gluings [C4], components [C5]; e [C34]	(4.1); components by Lemma B.1	the 66 components also by the mass formula [C5]; rootless mass (§B.2); e recounted for $\lvert C\rvert \le 64$; the program for e in rank 24 (§5.6)
order 3	gluings and e [C8]	(4.1); components by mass and (B.2)	$3-(16,0)-0$ by [Hen13]; the program for e in rank 24 (§5.6)
$(16,16)$, other 10 pairs	gluings [C4]; e [C34]	(4.1); components by Lemma B.1	components by a classification of the genera (§B.2); the program for e in rank 24 (§5.6)
M_3	formula of App. C [C35]	— (a formula)	none at rank 8; the same formula agrees at ranks 2, 4, 6, and with $2.\mathrm{Co}_1$ at rank 6
orders 5, 7, 17, 31	gluings and e [C9]	(4.1); components by Table D1 and (B.2)	$17-(2,0)-0$ by [Jür15]; App. D against the Leech lattice (§D.3); the program for e in rank 24 (§5.6)
M_4	formula of App. C [C35]	— (a formula)	a list of 514 classes
M_5	formula of App. C [C35]	— (a formula)	an explicit count of the 16 types of rank 2 (Table C8)

The historical checks involving higher cyclotomic masses concern the auxiliary assignment calculation. The classification-based total uses only the inputs of §5.2 of the paper. Agreement of the two arithmetic organizations cannot prove a structured-pair list complete if a class is absent from both.

5.5 Checks on known cases

The tools of §4 and of Appendices B–D were run on cases whose answers are known, and Table 7b lists these checks. They are not part of the proofs; they show that the programs reproduce published values and classical classifications wherever such values exist. The checks of one value by a second route inside this paper are in Table 7a and are not repeated here. The tests and scripts are [C37].

Table 7b. Checks of the tools on known cases [C37]. All agreements are exact unless a standard error is given.

tool	compared with	range	result
mass formula of Minkowski–Siegel	[SPLAG, Table 16.4]; the classes of ranks 8, 16, 24 with their $\lvert\mathrm{Aut}\rvert$ ([SPLAG, Tables 16.1, 16.6, 16.7])	ranks 8, 16, 24, 32	agree; the 24 Niemeier contributions agree entry by entry

tool	compared with	range	result
King's table [Kin03]	the Minkowski–Siegel mass of rank 32 and Siegel's average number of roots 16,320 / 3,617	all 13,218 rows	agree; this checks the table for internal consistency only
rootless mass m [C14]	King's value [Kin03, Table 1]; the known values in ranks 8, 16, 24	all root systems of rank at most 32	agree; the method is King's, so this checks our implementation
rootless masses of 2-elementary genera (§B.2)	the Niemeier lattices and the Leech lattice [SPLAG, Table 16.1]; for $(24, 2, \delta)$ a direct count of sublattices of index 2 of the Niemeier lattices	rank 24	agree
unimodular lattices without vectors of norm 1 (start of Lemma B.1)	numbers [SPLAG, Table 2.2], numbers of roots [SPLAG, Table 16.7], masses [SPLAG, Table 16.2], $\lvert\mathrm{Aut}\rvert$ / $\lvert W(R)\rvert$ [SPLAG, Table 17.1]	ranks at most 24	agree
existence of 2-elementary genera	[Nik79, Thm 3.6.2]	ranks at most 32, all s, δ	no disagreement
random gluings of involution components	exhaustive enumeration in ranks at most 5; the average number of roots 4,080 /691	samples of 2,000 per case	agree within 1.91 standard errors
lists of involution gluings	estimates of M from random gluings	types $(16, 16)$, $(8, 24)$	within 0.44 standard errors for $(16, 16)$; within the error for $(8, 24)$
type $(8, 24)$ (Theorem B.3)	the counts of the 7-design equations $N_0, ..., N_3 = 12{,}240, 122{,}880, 11{,}520, 0$	all 14,482 lattices K	agree; each K has average 80 roots over its 270 Niemeier lattices
images of the 120 root lines of E_8 in $E_8/3E_8$ (type $3-(12,8)-8$)	their number $\lvert\mathrm{GO}_8^+(3)\rvert$ / $\lvert W(E_8)\rvert = 56{,}862$	all images under $\mathrm{GO}_8^+(3)$	agree
rootless masses over $\mathbb{Z}[\zeta_p]$ (Appendix D)	the Leech lattice and the character table of $2.\mathrm{Co}_1$	$p = 5, 7, 13$	agree (§D.3)
elements of order 3 without fixed vectors	[Hen13, Thm 1, Table 4]	the 80 classes of the genus	the table, as transcribed, sums to the mass of [Hen13, Thm 1]; our eleven lattices give its eleven orders $\lvert\mathrm{U}\rvert$
elements of order 17	[Jür15, Satz 3.6.9]	the four lattices	agree (§4.2 of the paper)
rootless Hermitian mass over $\mathbb{Z}[i]$ (Appendix C)	the classes of ranks 4 and 8; a complete enumeration of 41 small types	ranks 4, 8	agree
bliss and `qfauto` in rank 32	$\lvert\mathrm{Aut}\ \mathrm{BW}_{32}\rvert = 48{,}126{,}558{,}103{,}142{,}400$; the orders predicted by [KN93] for its twelve lattices; isometries between lattices from different chains	16 lattices	agree; generators and isometries checked in integers

tool	compared with	range	result
the codes of [KN93]	[KN93, Thm 1] and the twelve codes of [KN93, Table 1]; $\sum 1/\lvert\mathrm{Aut}\ C_0\rvert$ against the value forced by King's mass	12 codes	one-to-one; for the column $\lvert\mathrm{Aut}\ C\rvert$ see the notes to Table A1

Not checked against known values. King's table was checked for internal consistency, not row by row against an independent computation of the other 13,217 root systems. The lists of the genera $(20, 6)$ over $\mathbb{Z}$ and $(13, 5)$ over $\mathbb{Z}[\omega]$ for elements of order 3 are complete by local formulas (§D.4) with no second route. The mass M_3 at rank 8 has no value to compare with (§5.4). Together with the last paragraph of §5.4, these are the places where the values rest on our programs alone.

5.6 The method in rank 24

In rank 24 the answer is known. The Leech lattice Λ is the only even unimodular lattice of rank 24 without vectors of norm 2 [SPLAG, Table 16.1], so the analogue of N is 1, and $\mathrm{Aut}\ \Lambda = \mathrm{Co}_0$. Every term of the analogue of (3.3) is therefore a sum over conjugacy classes of Co_0, and can be read from the character table of $2.\mathrm{Co}_1$ in GAP. Table 7c compares these values with the values given by the tools of §4 and of Appendices C and D, run in rank 24 [C38].

This is a check of the tools, not an independent proof. We did not redo §3 in rank 24: the types of elements were taken from the classes of Co_0, and the comparison uses the uniqueness of Λ. For the involutions and for nine types of odd prime order, the contributions E including the numbers $e(h)$ are compared. The programs that count $e(h)$ [C34, C8, C9] were used unchanged; the scripts for rank 24 replace only the places of the outputs and the test of extremality of each gluing, which is not automatic in rank 24. Not included are the contribution E of the type $3-(12,0)-0$ (where $C(h) \cong 6.\mathrm{Suz}$; only its rootless mass is compared), the types $11-(2,4)-2$ and $23-(1,2)-1$, and a version in rank 24 of the reduction of types in §3. Measured by the values from Co_0, the terms compared in Table 7c add up to 0.693 of the total 1.

Table 7c. The tools in rank 24 against the classes of $\mathrm{Co}_0 = \mathrm{Aut}\ \Lambda$ [C38]. In every row the value computed by the tool equals, as a fraction, the value read from Co_0: $1/\lvert\mathrm{Co}_0\rvert$, the sum of $1/\lvert C(g)\rvert$ over the classes of the given kind, the order of the centralizer of an involution, or the contribution $E = \sum e(h)/\lvert C(h)\rvert$ of a type.

term in rank 24	computed by	value
rootless mass m of rank 24	King's inclusion–exclusion, as for m in rank 32 [C14]	$1/\ 8{,}315{,}553{,}613{,}086{,}720{,}000 = 1/\lvert\mathrm{Co}_0\rvert$
M_2 ($\mathbb{Z}[i]$, rank 12)	the formula of Appendix C	$1/\ 2{,}012{,}774{,}400$
M_3 ($\mathbb{Z}[\zeta_8]$, rank 6)	the formula of Appendix C.8	$1/\ 48{,}384$
M_4 ($\mathbb{Z}[\zeta_{16}]$, rank 3)	Lemma 2.4: the rank is odd, so there is no such lattice	0
order 3, no fixed vectors ($\mathbb{Z}[\omega]$, rank 12)	rootless mass of Appendix D (§D.4)	$1/\ 2{,}690{,}072{,}985{,}600$
order 5, no fixed vectors ($\mathbb{Z}[\zeta_5]$, rank 6)	rootless mass of Appendix D; $e(h)$ by the programs of [C9], unchanged	$1/\ 6{,}048{,}000$; $E = 191/\ 3{,}375$
order 7, no fixed vectors ($\mathbb{Z}[\zeta_7]$, rank 4)	as above	$1/\ 35{,}280$; $E = 92/\ 2{,}205$
order 13, no fixed vectors ($\mathbb{Z}[\zeta_{13}]$, rank 2)	as above	$1/312$; $E = 1/39$
involutions of type $(16, 8)$	gluings as in §4.1 of the paper: one double coset,	$\lvert\mathrm{Stab}(\varphi)\rvert = 178{,}362{,}777{,}600$; $E = 465{,}333{,}121\ /\ 178{,}362{,}777{,}600$

term in rank 24	computed by	value
	extremal, giving Λ; $e(\sigma)$ by the program of [C34], unchanged	
involutions of type $(8,16)$	the same double coset seen from $-\sigma$; $e(-\sigma)$ as above	$E = 5{,}721{,}075{,}841$ / $178{,}362{,}777{,}600$
involutions of type $(12,12)$	gluings as in §4.1 of the paper: six double cosets, one extremal, giving Λ with $\sigma \sim -\sigma$; e as above	$\lvert\mathrm{Stab}(\varphi)\rvert = 389{,}283{,}840$; $E = 5{,}069$ / $77{,}856{,}768$
type $5-(4,8)-4$	gluings and $e(h)$ as in §4.2 of the paper, by the programs of [C9], unchanged; one pair, giving Λ	$E = 38/\ 1{,}125$
type $5-(5,4)-3$	as above	$E = 97/\ 1{,}875$
type $7-(3,6)-3$	as above	$E = 8/147$
type $3-(6,12)-6$	as above, with the programs for order 3 [C8]	$E = 641/\ 6{,}561$
type $3-(8,8)-8$	as above	$E = 26/243$
type $3-(9,6)-5$	as above	$E = 3{,}736$ / $19{,}683$

For $p = 13$ the walk by neighbours in the genus $(2,2)$ did not close; the list of rootless lattices in the genus $(2,0)$ was obtained from it by the step of §B.4 from $(z, s+2)$ to (z,s), and it is complete because its mass equals the value $1/312$ of the mass formula; identity (B.2), which would need the complete list in $(2,2)$, is not used.

The row of M_4 is 0 by Lemma 2.4, not by formula (C.2). Our implementation of (C.2), applied outside its range to the odd rank 3, returns a negative number: it does not impose the condition that the rank t of a complement and the number $d(C)$ of generators of the 2-part of its discriminant group have the same parity. Appendix C.8 uses (C.2) only for even ranks, and for the three cases used in this paper, $(a,r) = (3,8),(4,4),(5,2)$, no term of the computation violates that condition [C38].

C.8 The levels $a = 3, 4, 5$

Let $3 \le a \le 5$, $\zeta = \zeta_{2^a}$, $m = 2^{a-1}$, $\mathcal{O} = \mathbb{Z}[\zeta]$, $\mathbb{K}^+ = \mathbb{Q}(\zeta+\zeta^{-1})$ with $d_{\mathbb{K}^+} = [\mathbb{K}^+ : \mathbb{Q}] = 2^{a-2}$, $\varpi_a = 1-\zeta$ (as in §C.3) and $\varpi_{\mathbb{K}^+} = \varpi_a\overline{\varpi_a}$, a uniformizer of $\mathbb{K}^+$ at 2. For $g \in \mathrm{Aut}\ L$ with $g^m = -1$ put $H(x,y) = \sum_{k=0}^{m-1}(x, yg^k)\zeta^k$ (for $a = 2$ this is the form h of §C.1). It is $\mathcal{O}$-valued and Hermitian with $(x,y) = \mathrm{Tr}_{\mathbb{K}/\mathbb{Q}}H(x,y)/m$; since the different of $\mathbb{K}/\mathbb{Q}$ is $m\mathcal{O}$, L is unimodular if and only if H is, and L is even if and only if $H(x,x) \in \mathrm{Tr}_{\mathbb{K}/\mathbb{K}^+}(\mathcal{O}) = \varpi_{\mathbb{K}^+}\mathcal{O}_{\mathbb{K}^+}$ for all x (by $(x, xg^{m-k}) = -(x, xg^k)$, $H(x,x) = (x,x) + \sum_{0<k<m/2}(x,xg^k)(\zeta^k+\zeta^{-k})$, and $\mathrm{Tr}_{\mathbb{K}/\mathbb{K}^+}(\mathcal{O}) = 2\mathcal{O}_{\mathbb{K}^+} + (\zeta+\zeta^{-1})\mathcal{O}_{\mathbb{K}^+} = \varpi_{\mathbb{K}^+}\mathcal{O}_{\mathbb{K}^+}$). These lattices, of rank $r = 32/m = 8,4,2$, play the role of theta lattices; a root is a vector with $(x,x) = 2$. Everything in §C.1–C.7 goes through with the following changes.

The genus. For odd r there are no such lattices (Lemma 2.4). For even r these lattices form one genus, whose mass is $\mu_r = 2^{1-rd_{\mathbb{K}^+}}\prod_{j=1}^{r}|L_{\mathbb{K}^+}(1-j,\chi^j)|$, χ the character of $\mathbb{K}/\mathbb{K}^+$: this genus is the genus $G_r^{\mathbb{K}}$ of [Jür15, §3.3.1] (locally hyperbolic at the prime above 2 and $\langle 1,\dots,1\rangle$ elsewhere), and the mass is [Jür15, Satz 3.3.15].

Root lattices (Proposition C.1). Let C be an irreducible component of the root system and 2^{a-b} the length of its orbit under g; the stabilizer acts on C through an element g' with $g'^{2^{b-1}} = -1$ (for $b \ge 1$). For $b = 1$ the orbit gives $C \otimes \mathcal{O}$. For $b \ge 2$, as in the proof of Proposition C.1, $C = D_n$ with g' a product

of negative 2^{b-1}-cycles, giving the lattice $G_k^{(b)}$ of §C.3 induced to $\mathcal{O}$, or $C = D_4$, $b = 2$, or $C = E_8$ with $b = 2$ (E_8^g) or $b = 3$: the elements of $W(E_8)$ with characteristic polynomial Φ_8^2 form one conjugacy class with centralizer of order 192 [C11]. $b \geq 4$ is impossible for E_8: for $b = 4$ it would make E_8 an even unimodular lattice of rank 1 over $\mathbb{Z}[\zeta_{16}]$, contradicting Lemma 2.4, and for $b \geq 5$ the degree 2^{b-1} exceeds 8. Proposition C.3 and the base-change paragraph after it were stated for $\kappa \leq 2$, but hold for every κ with the same proofs once $G_k^{(\kappa)}$ carries the normalized trace form $(x, y) = 2^{1-\kappa} \operatorname{Tr} \sum_p x_p \overline{y_p}$, which is the restriction of $\operatorname{Tr}_{\mathbb{K}/\mathbb{Q}} H/m$: a root of unity $\eta \neq \pm 1$ of 2-power order has trace 0, so the half-edges $(1 - \eta)e_p$ and the edges have norm 2 while $2e_p$ has norm 4; every root of unity is $\equiv 1$ modulo ϖ_κ; $v_{\varpi_\kappa}(1 - \eta) = 2^{\kappa - j}$ for η of order 2^j; and the copies $\zeta^j C$ in $C \otimes_{\mathcal{O}_b} \mathcal{O}_\kappa$ are orthogonal because $\operatorname{Tr}_{\mathbb{K}_\kappa/\mathbb{K}_{\max(b,1)}} \zeta^j = 0$ for $0 < j < 2^{\kappa - \max(b,1)}$. So the coefficients $c(T)$ are obtained as before; for $G_1^{(3)}$ the coefficient is 0. For $C = E_8$ with $b = 3$, the root sublattices over $\mathcal{O}_3$ were enumerated in the model of E_8 on which g acts by two negative 4-cycles of the coordinates: besides 0 and E_8 they are of the types A_1 (18 of them), A_2 (8), G_2 (4), $G_1^{(3)}$ (6), $G_2^{(3)}$ (3), A_1^2 (9), $A_1 \perp G_1^{(3)}$ (6) and $G_1^{(3)} \perp G_1^{(3)}$ (3), where A_1, A_2 and G_2 stand for their base changes to $\mathcal{O}_3$ (so $G_1^{(2)} \otimes \mathcal{O}_3 = A_1 \otimes \mathcal{O}_3$ is among the 18); with $c(A_1) = -1$, $c(A_2) = c(G_2) = 2$, $c(A_1^2) = 1$ and $c\left(G_1^{(3)}\right) = c\left(G_2^{(3)}\right) = 0$ (the last by Proposition C.3 for $\kappa = 3$, $k = 2$) this gives $c = -16$ [C35].

The Siegel-type formula (Theorem C.4, Lemma C.5). The proof is the same with $\mathbb{Q}$ replaced by $\mathbb{K}^+$: the measure at infinity is the product over the $d_{\mathbb{K}^+}$ real places, and Weil's comparison of Tamagawa numbers holds over any number field [Wei65, n°53–54]. Proposition C.7 holds with the same proof at the prime above 2, whose residue field is $\mathbb{F}_2$, and at an odd prime $\mathfrak{p}$ of $\mathbb{K}^+$ with p^{-j} replaced by $N(\mathfrak{p})^{-j}$ and $\chi(p)$ by $\chi(\mathfrak{p})$. In Theorem C.8,

$$S'_t = 2^{1 - t d_{\mathbb{K}^+} - t[t \text{ odd}]} \prod_{j=1}^{t} |L_{\mathbb{K}^+}(1 - j, \chi^j)|,$$

and $f_\mathfrak{p}$ is defined by the same sums. The constant is obtained as in the proof of (C.6). At infinity the measure is the product over the real places v of the densities of Lemma C.5, divided by the covolumes of the integral lattices: $\mathcal{O}$ has covolume $2^{-d_{\mathbb{K}^+}} \sqrt{|D_\mathbb{K}|} = 2^{1 - d_{\mathbb{K}^+}} D_{\mathbb{K}^+}$ in $\mathbb{K} \otimes \mathbb{R} = \mathbb{C}^{d_{\mathbb{K}^+}}$ (the relative discriminant of $\mathbb{K}/\mathbb{K}^+$ has norm 4, so $|D_\mathbb{K}| = 4 D_{\mathbb{K}^+}^2$), and Λ_n, with diagonal in $\varpi_{\mathbb{K}^+} \mathcal{O}_{\mathbb{K}^+}$, has covolume $\left(2\sqrt{D_{\mathbb{K}^+}}\right)^n \left(2^{1 - d_{\mathbb{K}^+}} D_{\mathbb{K}^+}\right)^{n(n-1)/2}$. Hence (C.3) holds with

$$\alpha_\infty(T) = c^{d_{\mathbb{K}^+}} N_{\mathbb{K}^+/\mathbb{Q}}(\det T)^{r-n} \left(2\sqrt{D_{\mathbb{K}^+}}\right)^n \left(2^{1 - d_{\mathbb{K}^+}} D_{\mathbb{K}^+}\right)^{n(n-1)/2 - rn}.$$

The functional equation over $\mathbb{K}^+$ gives $|L_{\mathbb{K}^+}(1 - j, \chi^j)| = \left(2(j-1)!(2\pi)^{-j}\right)^{d_{\mathbb{K}^+}} D_{\mathbb{K}^+}^{j - 1/2} 4^{(j-1/2)[j \text{ odd}]} L_{\mathbb{K}^+}(j, \chi^j)$. Inserting this and $\mu_r = S'_r$ into $\mu_r \alpha_\infty(T) \prod_{j=t+1}^{r} L_{\mathbb{K}^+}(j, \chi^j)^{-1}$, the factors π and $(j-1)!$ cancel as before, the powers of $D_{\mathbb{K}^+}$ cancel exactly ($n(2r - n)/2 + n^2/2 - rn = 0$), and the terms of the exponent of 2 proportional to $d_{\mathbb{K}^+}$ add up to $d_{\mathbb{K}^+}(n + t - r) = 0$ after comparison with $2^{-t d_{\mathbb{K}^+}}$ in S'_t. The remaining exponent is the number E of the proof of (C.6), which is 0. For $t = r$ the formula gives μ_r, in agreement with [Jür15, Satz 3.3.15].

2-adic Jordan decomposition (Lemma C.9). Over the completion $\mathcal{O}_{\varpi_a}$ of $\mathcal{O}$ at ϖ_a, all statements and proofs of Lemma C.9 hold with 2 replaced by $\varpi_{\mathbb{K}^+}$, $\mathbb{Z}_2$ by $\mathcal{O}_{\mathbb{K}^+, \varpi_{\mathbb{K}^+}}$, $1 + 4\mathbb{Z}_2$ by $1 + \varpi_{\mathbb{K}^+}^2 \mathcal{O}_{\mathbb{K}^+}$ and the unit 3 by $1 + \varpi_{\mathbb{K}^+}$. The facts used are $\mathcal{O}_{\mathbb{K}^+}/\varpi_{\mathbb{K}^+} = \mathbb{F}_2$, $2 \in \varpi_{\mathbb{K}^+} \mathcal{O}_{\mathbb{K}^+}$, $\overline{\varpi_a} \equiv \varpi_a \pmod{\varpi_{\mathbb{K}^+}}$, $\operatorname{Tr}_{\mathbb{K}/\mathbb{K}^+}\left(\varpi_a^k \mathcal{O}\right) = \varpi_{\mathbb{K}^+}^{\lfloor (k+2)/2 \rfloor} \mathcal{O}_{\mathbb{K}^+}$ (the relative different is $\varpi_a^2 \mathcal{O}$), and $N\left(\mathcal{O}_{\varpi_a}^\times\right) = 1 + \varpi_{\mathbb{K}^+}^2 \mathcal{O}_{\mathbb{K}^+}$: the inclusion holds because $N(w + s\varpi_a) = w^2 + \varpi_{\mathbb{K}^+}(ws + s^2) \equiv 1 \left(\bmod \varpi_{\mathbb{K}^+}^2\right)$ for a unit w and $s \in \mathcal{O}_{\mathbb{K}^+}$, and equality because the norm group has index 2 in the units of the ramified extension. In the proof of (iv), the norm of $e_1 + \lambda e_2$,

$\lambda = w + s\varpi_a$, divided by $\varpi_{\mathbb{K}^+}^{e+1}$ is $a + bw^2 + w$ modulo $\varpi_{\mathbb{K}^+}$, again with derivative 1 in w. The dynamic programme of §C.6 then runs unchanged.

Table C8. Rootless masses for $a = 3, 4, 5$ [C35].

a, r	root types	non-zero	$M_a = \mu_r^{\emptyset}$	μ_r	independent check
3, 8	1,270	174	1,373,459,063 / 7,741,440	34,773,704,473,453 / 112,742,891,520	none at $r = 8$; at $r = 6$ the rootless mass 1/ 48,384 is $1/\ \lvert C_{\mathrm{Co}_0}(g)\rvert$ for the unique class with $g^4 = -1$
4, 4	122	29	6,983 /640	11,979,143 / 786,432	list of 514 classes (238 rootless), both masses exact
5, 2	21	7	31/24	1,455 / 1,024	direct count of the 16 types of rank 2, using $h(\mathbb{Q}(\zeta_{32})) = 1$ [MM76, Main Theorem]

Checks. For $a = 3$ and $r = 2, 4, 6$, every root system with non-zero mass gives $1/m(R)$ equal to the order of the unitary group of the corresponding lattice (192 for $r = 2$; $192^2 \cdot 2$, $2 \cdot 46{,}080$ and $8^4 \cdot 4!/2$ for $r = 4$; at $r = 6$ the 17 root systems with non-zero mass map to Niemeier root systems or the Leech lattice), and the rootless mass 1/ 48,384 at $r = 6$ is the value from the character table of $2.\mathrm{Co}_1$. For $a = 4$, the ϖ_a-neighbours of the 514 classes close up, $\sum 1/\ \lvert\mathrm{U}\rvert = \mu_4$ and the rootless part is 6,983 /640 exactly, and the 29 root types with non-zero mass match the 29 non-zero values of (C.2) one by one. For $a = 5$, for each root lattice R of rank 2 the lattices with root system exactly R correspond to the g-stable Lagrangian submodules of $D(R)$ creating no new root, so $m(R) = \#\{C\}/\ \lvert\mathrm{U}(R)\rvert$; an explicit count over all submodules gives $\sum_R m(R) = 397/\ 3{,}072$ over the 16 types of rank 2, and $\mu_2 - 397/\ 3{,}072 = 31/24$, the value of (C.2), the formula giving mass 0 to the four types of rank 1 [C35].

Historical Table 4 with assignment weights

Table 4. Involution types. F and K are the components (number of classes, s, δ); "classes" and "extremal" count double cosets $[\varphi]$ and those giving an extremal L, that is, isometry classes of pairs (L, σ). $M = \sum_{\text{extremal}} 1/\ \lvert\mathrm{Stab}(\varphi)\rvert$ counts both orientations; E is the contribution to (3.3); $E = M$ exactly when $e = 1$ throughout. The pairs of type $(16, 16)$ are in Table 6 of the paper [C4, C5, C6, C34, C7].

type	F	K	classes	extremal	M (approx.)	E (approx.)
$(15,17)$	$\sqrt{2}A_{15}^{+}$; 1, $s=15$, $\delta=1$	1 class, $s=15$, $\delta=1$	82,086	20,502	9,664.170	9,664.170
$(14,18)$	$\sqrt{2}(E_7^2)^{+}$; 1, $s=14$, $\delta=1$	5 classes, $s=14$, $\delta=1$	54,649	16,590	6,876.012	6,948.066
$(12,20)$	$\sqrt{2}D_{12}^{+}$; 1, $s=12$, $\delta=1$	66 classes, $s=12$, $\delta=1$	27,434	13,798	5,507.962	5,815.972
$(8,24)$	$\sqrt{2}E_8$; 1, $s=8$, $\delta=0$	14,482 lattices, $s=8$	14,482	14,482	6,017.813	6,185.440
$(16,16)$	10 classes, $8 \le s \le 16$	same	233,544	46,805	9,883.635	10,077.337

Computation references

Table 11. Computations and code. References such as [C1] in the text point to the rows of this table.

ID	computation and code	CPU time

C1	The lattices of S_{gen} (Theorem B): the lattices reached by the entries of the lists of §4, their automorphism groups (bliss), conjugacy classes (GAP), all their entries, and the identification of equal lattices
	`certificates/general-part-identification.json`
	`research record: C1/`
		264 h

C2	Types of elements of odd prime order: existence of genera, rootless Hermitian masses, exact linear programs on theta series (PPL), and complete genus lists used to exclude all but the 19 types of Table 4
	`certificates/odd-prime-masses.json`
	`research record: C2/`
		—

C3	Six finite rational infeasibility certificates for the remaining prime-order types, with the bases of the theta series of the components fixed, every row checked exactly, and the bases and the signs of the discriminants checked
	`certificates/type-exclusion-lp.json`
	`research record: C3/`
		—

C4	Involution types $(d, 32-d)$ with $d \in \{12,14,15,16\}$: the gluing classes $[\varphi]$, their stabilizers, the extremality flag, and the mass identity $\sum 1/\ |\text{Stab}| = |O(q)|\ /(|\text{Aut } F|\ |\text{Aut } K|)$ for every pair of components
	`certificates/involution-gluing-masses.json`
	`research record: C4/`
		—

C5	Components of type $(12,20)$: the 66 classes of rank-20 2-elementary lattices with $s=12$, $\delta=1$ and minimum ≥ 4, by the mass formula and independently by chains of index-2 overlattices; isometry decisions by bliss canonical forms and norm-4 configurations
	`certificates/involution-gluing-masses.json`
	`research record: C5/`
		$\gtrsim$ 37 h

C6	Involution type $(8,24)$: the 14,482 rank-24 lattices K with $\min K \ge 4$ and $\min K^{\#} \ge 2$, via the reduction to Niemeier lattices (Theorem B.3), the orbits of good subspaces for the Leech lattice and A_1^{24}, and identity (B.1) for A_2^{12}
	`certificates/type-8-24-masses.json`
	`research record: C6/`
		—

C7	Components of involution types of ranks 8 to 20: chains of index-2 overlattices from $\sqrt{2}U$ (Lemma B.1); in rank 16 also the classification of the seventeen genera of even 2-elementary lattices by Kneser 3-neighbours, completed by the mass of the genus (a check, not used in the proofs)
	`certificates/involution-gluing-masses.json`
	`research record: C7/`
		—

ID	computation and code	CPU time
C8	Elements of order 3: the ten types $3-(z,f)-s$, their components, gluing classes and the numbers $e(h)$ certificates/odd-prime-masses.json research record: C8/	—
C9	Elements of order $p=5,7,17,31$: the nine types, their components and the numbers $e(h)$; the rootless masses (D.4)–(D.5) and their checks; the existence conditions of Proposition D.1 compared with the oddity formula; the terms $g(Z_+)$ of (B.2) for $p=5$, recomputed for every class with the functions of the list program certificates/odd-prime-masses.json research record: C9/	—
C10	The classes of $K_{12}^{\#}/K_{12}$: the 756 vectors of norm 4/3 lie three in each of the 252 classes with $q=4/3$ (a check of the argument of §4.2) research record: C10/	—
C11	The rootless mass M_2 of rank-16 Hermitian $\mathbb{Z}[i]$-lattices whose trace form is even unimodular (exact rational arithmetic) certificates/gaussian-rootless-mass.json research record: C11/	64 min
C12	The 76,742 classes of rootless Hermitian $\mathbb{Z}[i]$-lattices of rank 16; the sum of their weights equals the value of [C11] as a fraction (computed by two programs); for every class, rerunning the enumeration used for its key returns exactly 73,440 vectors of norm 4, no two equal up to sign, and D and the canonical keys recomputed from it equal the saved ones (the time given is that of this check alone; the time of the lists was not recorded) certificates/gaussian-list-weights.json research record: C12/	≥ 15.9 h
C13	Direct classification-based aggregation of the total from King's mass and the saved lattice table; the sources of the group orders of S_{gen}, and a check that no planned input was left out. No thirteen-term values, assignment counts or higher cyclotomic masses are read research record: C13/	—
C14	The mass $m=m_{32}(\emptyset)$ of rootless even unimodular lattices of rank 32 by King's inclusion–exclusion over root systems, reimplemented; agrees with the value tabulated in [Kin03] as a fraction certificates/king-rootless-mass.json research record: C14/	23 min
C15	King's exact value for the empty root system, and the value used in the paper; the primary mass normalization uses full automorphism orders certificates/king-rootless-mass.json research record: C15/	—
C16	Rootless masses of the Eisenstein genera (N,s,b) by (D.6), (D.8), (D.9) (Appendix D.4); the 105 classes of type $(13,5)$; the checks of Table D4 certificates/odd-prime-masses.json research record: C16/	—
C17	Rootless masses of the p-elementary $\mathbb{Z}$-genera $(20,6)$, $p=3$, and $(16,4)$, $p=5$, by (D.10)–(D.12); the 24 lattices F; the checks of Table D5 certificates/odd-prime-masses.json research record: C17/	—
C18	Finite proof of the elementary quadratic 2-form element orbits by actual isometries on support patterns, for all 230 diagonal forms of dimensions 1 to 20	—

ID	computation and code	CPU time
	certificates/quadratic-2-form-orbits.json research record: C18/	
C19	Full rank-16 root-type index and inclusion-exclusion sum, with regenerated exceptional coefficients and the local DP statistics of [C11] as inputs certificates/gaussian-rootless-mass.json research record: C19/	—
C20	Closed formulas for the root-sublattice coefficients checked against all 12,934 terms of [C19], and the classes with $g^2 = -1$ in $W(F_4)$, $W(E_8)$ and $2.\mathrm{Co}_1$ from the character tables of CTblLib (a method different from [C11]) certificates/root-coefficients.json research record: C20/	—
C21	Lemma 6.2 checked on all groups of order 4 to 190 with a central involution (supplementary; the lemma is proved in §6.2) research record: C21/	12 min
C22	Full automorphism groups of the two exceptional known lattices, recomputed from Gram matrices alone by the short-vector local-graph and integral-stabilizer algorithm; separate from the structural conjugacy-class calculation certificates/exceptional-full-groups.json research record: C22/	—
C23	39 positive identifications of unlabelled records by explicit unimodular integer isometries; separate from non-isometry decisions and signature invariance certificates/general-part-identification.json research record: C23/	—
C24	Full-group signatures of the unlabelled records compared with every successful record of the general part; the minimum-shell profiles of the 27 unlabelled lattices; the 67 integer isometries identifying the lattices with $\lvert\mathrm{Aut}\ L\rvert > 10^7$ (the signature report comes from the first version of the comparison program; the current version reproduces the same signatures and adds diagnostics) certificates/general-part-distinctness.json certificates/general-part-identification.json certificates/general-part-witnesses.json research record: C24/	—
C25	Read-only count of exact entry identifiers by type in all successful general-part records, and the identification classes containing an identifier of an exact type (12,519 of 12,546), with one witness row for each class extracted by a separate program and replayed from the saved rows; conditional on full groups, complete class rows and correct lookups certificates/general-part-distinctness.json certificates/general-part-witnesses.json research record: C25/	—
C26	The lattices of $S_{C_2^2}$ (Theorem B): the trivial involution entries not met by S_{gen} in [C1] certificates/c2xc2-part.json research record: C26/	3 min
C27	The lattices of S_{CQ} (Theorem B): cyclic and generalized quaternion groups; the 849 triples of anti-linear classes, each shown to be one lattice by two explicit isometries checked in integers certificates/quaternion-triples.json research record: C27/	≥ 22.5 h

ID	computation and code	CPU time

C28	Inverse-gluing partition for $S_{C_2^2}$, matching the entries to the lattices by their identifiers, and all 1,698 integer isometries connecting the 849 quaternion triples, using the Gram matrices of the Gaussian classes
	`certificates/c2xc2-part.json`
	`certificates/quaternion-triples.json`
	`research record: C28/`
		—

C29	Read-only consistency of the saved full-group records of the general part with the subtraction lemma: trivial involution entries and their pair identifiers, Gaussian keys in the final list with equal unitary orders and anti-linear flags, and the provenance of the common Gaussian key function; the subtraction recomputed as set differences of individual entries and keys from the saved rows
	`certificates/general-part-witnesses.json`
	`certificates/subtraction-records.json`
	`research record: C29/`
		—

C30	Identification of Aut L for the 12,536 lattices of S_{gen} with $\lvert\mathrm{Aut}\ L\rvert \le 10^7$ (Small Groups Library, structure description, chief series; presentations and conjugacy classes, checked against the generators; the order of every presentation, for nine groups by coset enumeration along a chain of point stabilizers); Table 9 and its checks against $\lvert S\rvert$, (5.1) and Corollary C (supplementary) `research record: C30/`	about 112 h
C31	Identity (5.1) from the classified lattices and its split by part (two implementations) `research record: C31/`	1 s
C32	The 26 previously known extremal lattices of rank 32 with $\lvert\mathrm{Aut}\ L\rvert > 2$ found before the comparison with the catalogue, and their location in the lists of §4 (supplementary) `research record: C32/`	1.35 h
C33	The extremal lattices of rank 32 of the catalogue [NS] and the lattice of [BB92] for $m = 51$: an isometry onto a lattice of Table A1 found with bliss, with $T^{\mathsf{T}}GT = G'$ and $\det T = \pm 1$ checked in integers (supplementary) `research record: C33/`	—
C34	The numbers $e(\sigma)$ for involution entries (lifting to $C(\sigma)$ and counting elements assigned to σ), and their recount inside Aut L for a subset of entries `research record: C34/`	42 h
C35	Rootless masses M_a over $\mathbb{Z}[\zeta_{2^a}]$ for $a = 3, 4, 5$ (auxiliary [Aux]); for $a = 4$ the list of 514 classes and its agreement with the formula; for $a = 5$ the direct count of the types of rank 2; the class of $g^4 = -1$ in $2.\mathrm{Co}_1$ `research record: C35/`	84 min
C36	The thirteen terms of the auxiliary count [Aux] read from the output files, and their sum (two implementations; auxiliary) `research record: C36/`	4 s
C37	Checks of the tools on known cases (auxiliary checks): masses, King's table, the unimodular lattices of rank at most 24, existence of genera, random gluings, the type $(8, 24)$, published classifications, automorphism groups `research record: C37/`	14 min
C38	The tools in rank 24 against the conjugacy classes of Co_0 (auxiliary checks), including the programs for e of [C34], [C8] and [C9] on the three involution types and nine types of odd prime order, and the range of the implementation of (C.2) in odd rank `research record: C38/`	—